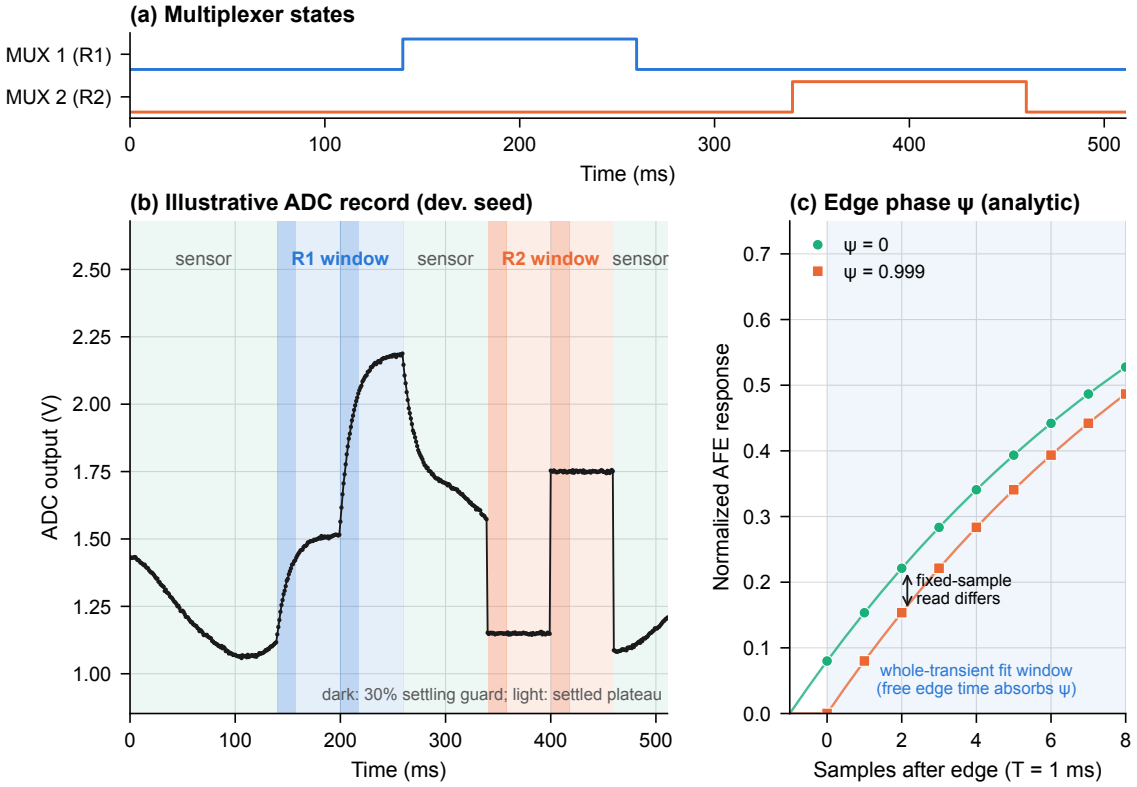
(a) Multiplexer states
MUX 1 (R1)
MUX 2 (R2)
0
100
200
300
400
500
Time (ms)
(b) Illustrative ADC record (dev. seed)
sensor
R1 window
sensor
R2 window
sensor
ADC output (V)
2.50
2.25
2.00
1.75
1.50
1.25
1.00
dark: 30% settling guard; light: settled plateau
0
100
200
300
400
500
Time (ms)
(c) Edge phase ψ (analytic)
ψ = 0
ψ = 0.999
Normalized AFE response
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
fixed-sample
read differs
whole-transient fit window
(free edge time absorbs ψ)
0
2
4
6
8
Samples after edge (T = 1 ms)